

KOMA - A Structural Equation Model in R

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Outline

- KOF Swiss Economic Institute
- Structural Equation Models (SEMs)
- KOMA R package
- KOMA Application to Financial Statistics

KOF Swiss Economic Institute



- Business Tendency Surveys
- KOF Economic Barometer
- ...
- Quarterly Forecasts for the Swiss Economy

Structural Equation Models



Why Structural Equation Model (SEM)?

- SEMs offer a straightforward interpretation of results, unlike VARs and DSGEs.
- Central banks and forecasting institutions continue to rely on large-scale SEMs.
- Existing software is outdated and requires an update.

A Small Macroeconomic Model

- Focus on theoretically motivated causal links, e.g., $x \rightarrow y$
- Consider the following model

consumption: $c_t = \alpha_0 + \alpha_1 gdp_t + \alpha_2 c_{t-1} + \varepsilon_{t1}$

investment: $i_t = \beta_0 + \beta_1 r_t + \varepsilon_{t2}$

gdp: $gdp_t = c_t + i_t + g_t$

- c_t , i_t , and gdp_t are **endogenous variables**
- r_t (interest rate) and g_t (government spending) are **exogenous variables**
- c_{t-1} are **predetermined variables**
- gdp_t is defined by an **identity equation**

KOMA R Package



Key Features of KOMA

- Bayesian estimation procedure (MCMC Methods)
 - Facilitates intuitive and straightforward density forecasting.
 - Allows for formal inclusion of prior knowledge in the analysis.
- User-friendly R-Package
 - Requires only text input for model specification.
 - Offers conditional forecasting capabilities similar to VARs.
 - Provides tools for both in-sample and out-of-sample model evaluation.
 - Includes dynamic plots and tables for reporting results.

Basic steps

1. Construct a string vector with all equations of the model:
 - Decide whether equations are stochastic or identity.
 - Declare all exogenous variables.
 - Choose informative priors or use non-informative priors (default).

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5. Compute forecasts using the `forecast` function.
6. Plot or print forecasts using `plot` or `print`.

Defining the Equations in KOMA



KOMA Syntax: Small Macro Model

$$c_t = \alpha_0 + \alpha_1 gdp_t + \alpha_2 c_{t-1} + \varepsilon_{t1} \quad \text{"consumption ~ gdp + lag(consumption,1)"}$$

$$i_t = \beta_0 + \beta_1 r_t + \varepsilon_{t2}$$

$$gdp_t = c_t + i_t + g_t$$

- Stochastic equations use $\sim$, identity equations use $==$.

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```
"consumption ~ gdp + lag(consumption,1)"
```

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```
"investment ~ interest_rate"
```

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KOMA Syntax: Small Macro Model

$c_t = \alpha_0 + \alpha_1 gdp_t + \alpha_2 c_{t-1} + \varepsilon_{t1}$ "consumption ~ gdp + lag(consumption,1)"

$i_t = \beta_0 + \beta_1 r_t + \varepsilon_{t2}$

"investment ~ interest_rate"

$gdp_t = c_t + i_t + g_t$

"gdp == 0.6*consumption + 0.2*investment + 0.2*government_spending"

- Stochastic equations use ~, identity equations use ==.

Lag Syntax

```
"consumption ~ gdp + lag(consumption,1)"
```

```
"investment ~ interest_rate"
```

```
"gdp == 0.6*consumption + 0.2*investment + 0.2*government_spending"
```

- KOMA accepts both `lag(variable,k)` and the shorthand `variable.L(k)`.
- Mixed lag requests are allowed, e.g. `x.L(1:2, 5)`.
- The parser expands lag ranges automatically and keeps the notation declarative.

Lag Syntax

```
"consumption ~ gdp + consumption.L(1)"
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Lag Syntax

```
"consumption ~ gdp + consumption.L(1) + consumption.L(2) + consumption.L(5)"
```

```
"investment ~ interest_rate"
```

```
"gdp == 0.6*consumption + 0.2*investment + 0.2*government_spending"
```

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Intercept Convention

```
"consumption ~ gdp + consumption.L(1)"
```

```
"investment ~ interest_rate"
```

```
"gdp == 0.6*consumption + 0.2*investment + 0.2*government_spending"
```

- Stochastic equations include an [intercept by default](#).
- Appending 0 removes the constant term, following standard R formula notation.
- Identity equations list only the fixed accounting relationship.

Intercept Convention

```
"consumption ~ 0 + gdp + consumption.L(1)"
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Priors

```
"consumption ~ gdp + consumption.L(1)"
```

```
"investment ~ interest_rate"
```

```
"gdp == 0.6*consumption + 0.2*investment + 0.2*government_spending"
```

- Coefficient priors are specified **inline within each equation** using braces of the form {mean, variance}.
- If no prior is given, KOMA uses the default non-informative setup.

Priors

```
"consumption ~ gdp + consumption.L(1)"
```

```
"investment ~ {-0.2,0.1} interest_rate"
```

```
"gdp == 0.6*consumption + 0.2*investment + 0.2*government_spending"
```

- Coefficient priors are specified **inline within each equation** using braces of the form {mean, variance}.
- If no prior is given, KOMA uses the default non-informative setup.

System of Equations

```
equations <- c(
  "consumption ~ gdp + consumption.L(1)",
  "investment ~ interest_rate",
  "gdp == 0.6*consumption + 0.2*investment + 0.2*government_spending"
)
exogenous_variables = c("interest_rate", "government_spending")
sys_eq <- system_of_equations(
  equations = equations,
  exogenous_variables = exogenous_variables
)
```

System of Equations

```
equations <- c(
  "consumption ~ gdp + consumption.L(1)",
  "investment ~ interest_rate",
  "gdp == 0.6*consumption + 0.2*investment + 0.2*government_spending"
)
exogenous_variables = c("interest_rate", "government_spending")
sys_eq <- system_of_equations(
  equations = equations,
  exogenous_variables = exogenous_variables
)
print(sys_eq)
#>
#> -- System of Equations -----
#> consumption ~ constant + gdp + consumption.L(1)
#> investment ~ constant + interest_rate
#> gdp == 0.6 * consumption + 0.2 * investment + 0.2 * government_spending
```

Extended Time Series Class



Extended Time Series Class `koma::ets`

- `koma::ets` extends `stats::ts` with metadata to move consistently between **levels** and **growth rates** — and back.

```
x <- ets(gdp, start = c(2019,1), frequency = 4,  
         series_type = "level", method = "diff_log")  
  
g <- rate(x)   # level -> growth rate  
x2 <- level(g) # growth rate -> level (reconstructed)
```

Extended Time Series Class `koma::ets`

- `koma::ets` extends `stats::ts` with metadata to move consistently between **levels** and **growth rates** — and back.

```
x <- ets(gdp, start = c(2019,1), frequency = 4,  
         series_type = "level", method = "diff_log")
```

```
g <- rate(x)    # level -> growth rate  
x2 <- level(g)  # growth rate -> level (reconstructed)
```

- Common operations — `window()`, `lag()`, `arithmetic`, `rebase()`, `concat()` — preserve the metadata.

Estimate and Forecast



Estimate

```
dates <- list(estimation = list(start = c(1990,1), end = c(2025,4)))  
estimates <- estimate(ts_data, sys_eq, dates)
```

Estimate

```
dates <- list(estimation = list(start = c(1990,1), end = c(2025,4)))  
estimates <- estimate(ts_data, sys_eq, dates)  
print(estimates)
```

Estimate

```
dates <- list(estimation = list(start = c(1990,1), end = c(2025,4)))
estimates <- estimate(ts_data, sys_eq, dates)

print(estimates)

#> -- Estimates -----
#> consumption ~ 0.34 + 0.03 * gdp + 0.08 * consumption.L(1)
#> investment ~ 0.57 - 0.01 * interest_rate
#> gdp == 0.6 * consumption + 0.2 * investment + 0.2 * government_spending
```

Estimate

```
dates <- list(estimation = list(start = c(1990,1), end = c(2025,4)))  
estimates <- estimate(ts_data, sys_eq, dates)  
summary(estimates)
```

Estimate

```
dates <- list(estimation = list(start = c(1990,1), end = c(2025,4)))
estimates <- estimate(ts_data, sys_eq, dates)
```

```
summary(estimates)
```

```
#> =====
```

```
#>                consumption      investment
```

```
#> -----
```

```
#> constant          0.34          0.57
```

```
#>                [ 0.27; 0.42]  [ 0.29; 0.85]
```

```
#> consumption.L(1)    0.08
```

```
#>                [-0.08; 0.26]
```

```
#> gdp                0.03
```

```
#>                [-0.01; 0.06]
```

```
#> interest_rate          -0.01
```

```
#>                [-0.25; 0.22]
```

```
#> =====
```

```
#> Posterior mean (90% credible interval: [5.0%, 95.0%])
```

ETH zürich Estimation period: 1996 Q1 - 2019 Q4

Forecast

```
dates <- list(forecast = list(start = c(2023, 1), end = c(2023, 4)))  
forecasts <- forecast(estimates, dates)
```

Forecast

```
dates <- list(forecast = list(start = c(2023, 1), end = c(2023, 4)))
forecasts <- forecast(estimates, dates)
```

```
print(forecasts)
```

```
#> attributes:
```

```
#>   series_type: list[5]
```

```
#>   method: list[5]
```

```
#>   anker: list[5]
```

```
#>
```

```
#> series:
```

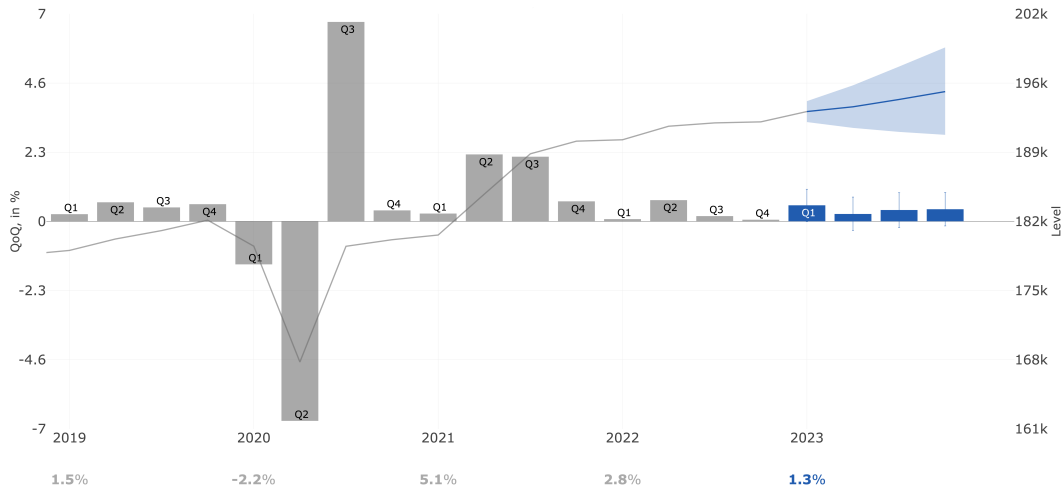
```
#>      consumption investment      gdp interest_rate government_spending
#> 2023 Q1      0.3733      0.6074 0.5497      1.1009      1.0213
#> 2023 Q2      0.3832      0.5512 0.2493      1.5227     -0.4545
#> 2023 Q3      0.3872      0.5518 0.3651      1.7075      0.1120
#> 2023 Q4      0.3870      0.6339 0.4003      1.7006      0.2064
```

Forecast

```
dates <- list(forecast = list(start = c(2023, 1), end = c(2023, 4)))
forecasts <- forecast(estimates, dates)

level(forecasts$mean$gdp)
#> attributes:
#>   series_type:  chr "level"
#>   method:      chr "diff_log"
#>
#> series:
#>           Qtr1      Qtr2      Qtr3      Qtr4
#> 2022                191668.9
#> 2023 192725.5 193206.5 193913.1 194690.8
```

Plots



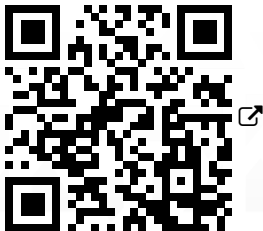
Conditional Forecasting

```
restrictions <- list(investment = list(value = weaker_investment, horizon = 1:4))
```

- Start from the baseline forecast for `investment`.
- Impose a weaker investment path over 2023 Q1–Q4.
- Recompute the forecast conditional on this restriction.
- The rest of the system adjusts endogenously.

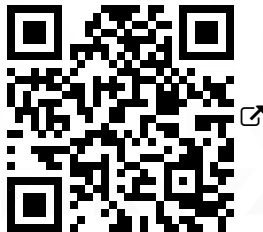
	Baseline GDP	Conditional GDP
2023 Q1	0.54	0.51
2023 Q2	0.25	0.22
2023 Q3	0.38	0.34
2023 Q4	0.41	0.36

KOMA R Package on GitHub



github.com/TimothyMerlin/koma

Documentation



timothymerlin.github.io/koma/

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